OpenMIBOOD: Open Medical Imaging Benchmarks for Out-Of-Distribution Detection

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Abstract

The growing reliance on Artificial Intelligence (AI) in critical domains such as healthcare demands robust mechanisms to ensure the trustworthiness of these systems, especially when faced with unexpected or anomalous inputs. This paper introduces the Open Medical Imaging Benchmarks for Out-Of-Distribution Detection (OpenMIBOOD), a comprehensive framework for evaluating out-of-distribution (OOD) detection methods specifically in medical imaging contexts. OpenMIBOOD includes three benchmarks from diverse medical domains, encompassing 14 datasets divided into covariate-shifted in-distribution, near-OOD, and far-OOD categories. We evaluate 24 post-hoc methods across these benchmarks, providing a standardized reference to advance the development and fair comparison of OOD detection methods. Results reveal that findings from broad-scale OOD benchmarks in natural image domains do not translate to medical applications, underscoring the critical need for such benchmarks in the medical field. By mitigating the risk of exposing AI models to inputs outside their training distribution, OpenMIBOOD aims to support the advancement of reliable and trustworthy AI systems in healthcare. The repository is available at https://github.com/remic-othr/OpenMIBOOD.

1. Introduction

The rapid proliferation of AI systems in everyday life raises concerns about their trustworthiness. According to the technical specification ISO/IEC TS5723:2022 [30], an important aspect of trustworthiness concerns a system's ability to handle unexpected inputs. Such inputs pose significant risks in safety-critical areas, including autonomous driving [11] and the healthcare sector [74]. Most AI systems are still trained and evaluated under the assumption that the training data distribution – referred to as in-distribution (ID) data – matches the distribution of data encountered post-deployment. However, this assumption overlooks the possibility of encountering data from unknown and unseen distributions, known as out-of-distribution (OOD) data. When confronted with such data, AI models often exhibit high confidence in their predictions, even when these predictions are entirely incorrect [18]. Such behavior can result in silent and potentially catastrophic failures, particularly in high-stakes domains like healthcare, where erroneous predictions could directly impact patient safety. To address this, OOD detection methods help distinguish ID from OOD inputs, allowing models to flag or discard unreliable predictions or refer them for human review. Since 2016, numerous OOD detection methods have emerged [72], but a unified, comprehensive benchmark was lacking. Yang *et al*. addressed this by introducing OpenOOD [72], an open-source codebase that integrates related fields and provides a consistent evaluation framework.

In their framework, OOD is divided into two categories. The more challenging near-OOD, characterized by exhibiting similar semantics or styles compared to the ID datasets, making it particularly difficult to distinguish them. And the less challenging far-OOD, typically featuring different semantic labels and styles [16, 72].

In a follow-up version, OpenOOD v1.5 [77], Zhang *et al*. expanded their framework to encompass large-scale and full-spectrum OOD detection [73], covering over 50 methods. In this full-spectrum setting, the authors merged ID with covariate-shifted in-distribution (cs-ID) data – data, retaining the same labels and class relationships as the ID data while exhibiting variations in the distribution of input features. They argue that this approach evaluates not only OOD detection capabilities but also the generalization performance of an AI system. While this approach has undeniable value, we argue for analyzing OOD detection and model generalization independently to yield more precise insights. Furthermore, depending on the application, distinguishing between ID and cs-ID can be just as critical. For instance, in medical imaging, where training data is often limited or inaccessible due to privacy constraints, achieving

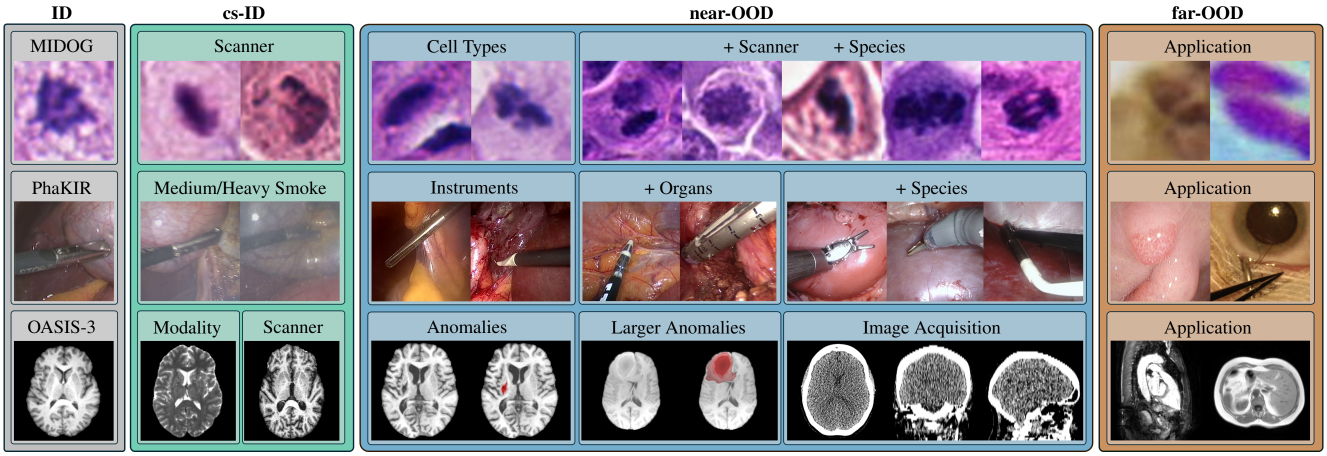

Figure 1. Illustration of all utilized datasets, categorized by varying degrees of domain shift, from cs-ID to near-OOD and far-OOD, presented from left to right: **MIDOG**: All distinct MIDOG domains [5], CCAgT [2, 4], FNAC 2019 [55]. **PhaKIR**: PhaKIR frames with smoke [53], Cholec80 [66], EndoSeg15 [10], EndoSeg18 [1], Kvasir-SEG [31, 50], CATARACTS [20]. **OASIS-3**: T2w modality and distinct scanner from OASIS-3 [38], ATLAS [43], BraTS [47], CT from OASIS-3 [38], MSD-H [64], CHAOS [32]. On top of each group the primary domain shift is presented. A '+' indicates that the previous domain shift is also included.

robust generalization may be challenging. In such cases, focusing on detecting cs-ID data can prove more impactful than solely aiming to enhance generalization performance.

We propose that, rather than combining ID and cs-ID data, OOD detection methods should be rigorously evaluated based on their capacity to differentiate between these two categories. This is particularly important in domains like healthcare, where the ability to identify subtle covariate shifts in data is essential to maintaining reliability.

To address the absence of medical images in the OpenOOD benchmark, we introduce three Medical Imaging Benchmarks (MIBs), which consist of 14 datasets organized into three distinct OOD evaluation settings named after their ID datasets: MIDOG [5], PhaKIR [53], and OASIS-3 [38]. An overview of these benchmarks and their involved domain shifts is provided in Fig. 1.

OOD detection methods are typically divided into two main types: post-hoc methods applied solely during inference, and methods requiring additional training steps. This work primarily focuses on post-hoc methods, valued for their ease of integration and model-agnostic nature. Besides these advantages, the analysis of Yang *et al*. showed that post-hoc methods are no worse than methods that require additional training steps [72].

Following an overview of the current benchmarks for OOD detection methods in the medical field (Sec. 2), we describe all evaluation settings, their underlying datasets, and the employed evaluation metrics (Sec. 3). Subsequently, we present the results of all evaluated methods, together with an in-depth discussion of these results (Sec. 4) and a summary of key findings and future research directions (Sec. 5).

Our main contributions are as follows:

- We introduce the most comprehensive OOD benchmarks for medical imaging across histopathology, endoscopy, and Magnetic Resonance Imaging (MRI) domains, evaluating 24 post-hoc methods and revealing gaps in current approaches for certain domain shifts in medical data.
- We show that selecting top-performing methods from benchmarks based on natural images likely leads to suboptimal performance on medical imaging data.
- To reproduce our results, we provide a comprehensive codebase including all evaluated OOD detection methods, along with download links for the employed machine learning models and links to all utilized datasets.

2. Related Work

OOD detection is central to medical image processing, as it ensures that deep learning models can appropriately handle and respond to data that falls outside their training distribution. Despite its importance, only a limited number of publications have systematically evaluated OOD methods in this specific domain [12, 68, 79]. Instead, most studies have prioritized the development of new methods [27].

The Medical Out-of-Distribution Analysis Challenge [79] serves as a major benchmark for evaluating OOD and anomaly detection in medical images. Its ID datasets include MRI scans of healthy adults and Computed Tomography (CT) scans from patients over 50 years old referred for colonoscopy. The OOD test data feature synthetic image corruptions, alterations, and destructions, aligning with the cs-ID category of [77], as well as images of unseen medical conditions that correspond to near-OOD. The challenge results reveal a substantial variance in the performance of OOD detection methods across different types of anomalies, indicating that none of the current approaches are sufficiently robust for immediate clinical deployment. However, this challenge has limitations: it does not comprehensively evaluate OOD detection methods across diverse medical imaging domains, nor does it offer a unified framework to guide and facilitate future research and development in the field.

Cao *et al*. [12] categorize OOD examples into three use cases: rejecting improperly preprocessed inputs (*i.e*. cs-ID), rejecting inputs that contain unseen conditions or artifacts (*i.e*. near-OOD), and rejecting unrelated inputs (*i.e*. far-OOD), such as cat pictures. Their evaluation spans eight post-hoc OOD detection methods and eight methods requiring additional training steps applied across three medical imaging contexts: chest X-ray, fundus imaging, and histology. However, the scope of their evaluation is notably limited, encompassing only basic post-hoc approaches. Additionally, the quality of the curated datasets presents challenges. For instance, evaluations involving natural images have limited relevance in a medical context. Furthermore, most evaluation settings rely on a single OOD dataset or even omit the essential context of near-OOD datasets.

In [68] the effectiveness of OOD detection methods is evaluated with a focus on medical 3D image segmentation. The ID datasets include CT images showing lung nodules and MRI images depicting benign brain tumors. Various OOD scenarios are evaluated, incorporating distribution shifts caused by different imaging devices and synthetic image corruptions (*i.e*. cs-ID), unseen diseases (*i.e*. near-OOD) and entirely different organs (*i.e*. far-OOD). The study evaluates three post-hoc methods and three methods that change the model architecture or training process. While the work contributes valuable insights into OOD detection within the context of 3D medical image segmentation, its scope remains limited. Notably, the study primarily emphasizes the evaluation of the author's data-centric OOD detection method.

In addition to studies focusing on the evaluation of various OOD detection methods, Hong *et al*. [27] offer a comprehensive review of the OOD detection landscape within medical image analysis. They argue that the differentiation between cs-ID and near-OOD from OpenOOD is not entirely applicable for medical images. For example, while a classifier trained on X-ray images might still perform accurately on low-contrast X-rays, its performance is likely to degrade significantly when applied to entirely different modalities, such as CT scans [27]. Under the strict taxonomy outlined in [72, 77], both shifts would technically fall under the cs-ID category, as class labels remain unchanged, despite the significant differences in modality.

To address these limitations, Hong *et al*. propose an alternative taxonomy, categorizing domain shifts into covariate, semantic, and contextual shifts. While this refinement offers greater precision, it does not account for certain semantic shifts, such as those observed across species in benchmarks like MIDOG and PhaKIR (Sec. 3). Consequently, this work adopts the established taxonomy of [72, 77] with a key modification: datasets with contextual differences from the ID data are classified as near-OOD if these differences are substantial enough to reasonably anticipate unreliable classification performance. This approach strikes the balance between the established taxonomy and the need for practical applicability in diverse medical imaging contexts.

3. OOD: Benchmarks and Metrics

To ensure reliable performance in machine learning models, robust control over their inputs is essential. OOD detection methods are designed to identify inputs that deviate significantly from a model's training distribution, as such unfamiliar inputs can lead to unreliable predictions. The primary objective of OOD detection is to flag instances that fall outside the data distribution used during training. However, many existing OOD detection methods rely heavily on classification predictions, necessitating a highly effective underlying classification model for each ID dataset. The following subsection introduces the classification tasks and provides an overview of the datasets associated with them, highlighting their relevance to the evaluation of OOD detection approaches.

3.1. Medical Imaging Benchmarks

This work builds upon the foundation of the existing OpenOOD benchmark [72–74, 77], which was originally developed for natural images. While following a similar structure and taxonomy, we extend this framework to the medical imaging domain by introducing OpenMIBOOD. To achieve this, we selected three datasets from diverse medical imaging fields as ID and curated corresponding cs-ID, near-OOD, and far-OOD datasets (Fig. 1). To ensure robust evaluation, all OOD datasets were further divided into validation and test sets. The validation sets are used for hyperparameter tuning, while the test sets were exclusively reserved for assessing OOD detection performance. Additional information on all datasets and their respective splits can be found in Appendix A.

MIDOG The MIDOG Challenge dataset [5] by Aubreville *et al*. comprises 503 images of Hematoxylin & Eosin stained histological whole slides. These images are organized into ten distinct domains, labeled as $1_{\text{a–c}}$, 2–5, $6_{\text{a,b}}$, and 7 (Appendix A.1), each exhibiting covariate shifts, semantic shifts, or both. These shifts arise from variations introduced by differing imaging hardware and staining protocols, as well as seven different cancer types from both human and canine species.

The dataset contains annotations for mitotic cells and mitotic cell lookalikes (imposters) in the form of 50×50 px regions. To adapt the original object detection challenge into a three-class classification task, we utilized the provided annotations to create separate crops for mitotic cells and imposter cells. Moreover, we created additional crops by shifting each annotation window 100 px to the right, such that these crops do not belong to either of the original categories. These additional crops were subsequently used as training data for the third class. To ensure the quality and consistency of the dataset, crops smaller than 50×50 px, typically occurring at the image borders, were excluded. Further, all additional crops overlapping with any annotation were also removed. The same procedure was applied to the remaining MIDOG dataset domains as well.

Since the primary distinction between the ID data and the data from domains 1_b and 1_c lies in the imaging devices used, we designated them as cs-ID. The remaining domains of the MIDOG dataset were classified as near-OOD because they exhibit semantic shifts and, in most cases, additional covariate shifts.

For the far-OOD category, we incorporated two datasets associated with distinct medical applications. (1) **CCAgT dataset** [2, 4]: This dataset contains images of cervical cancer cells stained using the AgNOR technique, which specifically highlights regions within the cell nuclei. We utilized the available annotations of cell nuclei and extracted 50×50 px crops centered on each nucleus. (2) **FNAC 2019 dataset** [55]: This dataset features Pap-stained images of breast FNAC samples. The crop extraction involved a binary threshold segmentation step, followed by a series of morphological operations to isolate cell clusters. From the processed images, the ten largest clusters were identified, and 50×50 px crops were extracted at the centroids of these clusters.

PhaKIR The PhaKIR-Challenge dataset, introduced by Rueckert *et al*. [53], consists of eight publicly available real-world human cholecystectomy videos collected from three hospitals. It includes annotations for instance segmentation and keypoint detection of 19 distinct surgical instruments, as well as for intervention phases. To maintain consistency and to avoid introducing unintended covariate shifts, only the first six videos, containing a subset of ten surgical instruments and originating from a single hospital, are included in this study, while two videos from other hospitals are excluded. For this benchmark, we train a classification model to identify surgical instruments. Using the provided annotations, we selected 2769 frames containing only a single surgical instrument to simplify the classification task. Further, 1891 images that do not contain any instruments are also selected and classified under the category ”No-Instrument”.

To create separate ID and cs-ID datasets from the PhaKIR-Challenge data, extracted frames were categorized into three smoke levels – none, medium, and heavy – based on the annotations from [54], reflecting the extent of smoke produced by tissue coagulation. Frames with no visible smoke were designated as the ID dataset, while frames with medium and heavy smoke were assigned to the cs-ID datasets. This separation ensures a clear distinction between the ID and the cs-ID datasets, enabling a robust evaluation of OOD detection performance with respect to covariate shifts caused by varying levels of smoke.

The following datasets were selected as near-OOD datasets, each introducing progressively greater semantic and covariate shifts. (1) **Cholec80 dataset** [66]: Although also featuring cholecystectomies, the instruments used differ from those in the ID dataset. Only those frames containing a single instrument were selected. Additionally, strong black vignettes present in almost all videos were removed by cropping a rectangular region within the vignette, preserving the original aspect ratio of the ID images. (2) **Instrument segmentation and tracking dataset (EndoSeg15)** [10]: From the Endoscopic Vision Challenge (EndoVis) 2015, this dataset includes frames from colorectal surgeries. We utilized images depicting rigid instruments that were distinct from those included in the ID dataset. (3) **Robotic scene segmentation dataset (EndoSeg18)** [1]: As part of the EndoVis Challenge 2018, this dataset features robotic instruments used in porcine nephrectomy procedures. To reduce the computational load, we utilized only the official test split from this dataset. Further, only the left frames from the stereo camera setup were used to maintain consistency with the other datasets. In contrast to the ID and Cholec80 datasets, we did not discard frames containing multiple instruments in EndoSeg15 and EndoSeg18.

For far-OOD evaluation, two datasets from entirely different medical fields were utilized: (1) **Kvasir-Seg dataset** [31]: This dataset consists of 1000 endoscopic images of polyps inside the bowel. (2) **CATARACTS dataset** [20]: Containing microscope videos of cataract surgeries, this dataset features a variety of ophthalmological instruments entirely different from those in the ID dataset.

OASIS-3 The OASIS-3 [38] dataset comprises 2842 primarily high-resolution longitudinal MRI and low-dose CT scans from 1378 subjects, covering modalities such as T1-weighted (T1w), T2-weighted (T2w), and FLAIR. The dataset includes individuals ranging from cognitively normal (CN) to various stages of cognitive decline, up to Alzheimer’s disease (AD). For this work, we specifically selected CN and AD subjects to facilitate a clearer distinction in brain tissue morphology. To ensure high-quality labels, we included only those MRI scans for which a clinical diagnosis is available within a 365-day window prior to the MRI acquisition date or a 182-day window after. If no diagnosis was available within this constraints, the last one to three prior diagnoses were reviewed and accepted, provided they consistently indicated AD. All selected scans were preprocessed by resampling to an isotropic voxel spacing of 1 mm^3 and applying skull-stripping using HD-BET [29]. For the ID dataset, only T1w MRI scans were used to train a classification model to distinguish between CN and AD subjects. To evaluate the effects of covariate shifts, we created two cs-ID datasets: (1) **Modality-based covariate shift**: The first set comprises T2w MRI scans, introducing a covariate shift due to a change in imaging modality. (2): **Scanner-based covariate shift**: All T1w scans acquired using a Siemens Vision device were withheld to form the second cs-ID dataset, representing a shift in the MRI scanner.

The near-OOD category includes datasets that introduce varying degrees of domain shifts, ranging from subtle to more pronounced differences: (1) **ATLAS dataset** [43]: This dataset introduces a semantic shift as it contains images with stroke lesions, which differ from the CN and AD patterns in the ID dataset. Only T1w scans were selected, resampled, and skull-stripped to ensure consistency with the ID image properties. We excluded lower-quality scans to avoid introducing unintended covariate shifts (Appendix A.3). (2) **BraTS Challenge dataset** [6, 7, 47]: Featuring large gliomas that affect substantial regions of the brain, this dataset presents a more significant semantic shift compared to ATLAS. As with ATLAS, only T1w images were included, and no preprocessing was required. (3) **CT scans from OASIS-3**: These scans represent a unique near-OOD category. Although they depict the same subjects as the MRI scans, technically retaining the CN and AD labels, the acquisition method differs fundamentally. While CT scans are more sensitive to calcifications [78], atrophy is less visible [41], with both being relevant for the classification task. These differences can result in extracted features that may not align with those from MRI scans. To ensure only regions containing semantic shifts are evaluated, we discarded scans for both the ATLAS and the BraTS Challenge datasets where the pathological area lies outside the region of interest – a 128 mm^3 volume centered in the brain.

For the far-OOD setting, two datasets from entirely different anatomical regions were selected to introduce significant semantic and structural differences: (1) **Medical Segmentation Decathlon Heart (MSD-H) dataset** [3, 56, 64]: The MSD-H dataset includes high-resolution contrast-enhanced MRI scans of the heart captured during a single cardiac phase. (2) **CHAOS Challenge dataset** [32, 33]: We utilize the in-phase sequences from this dataset, which contains dual-echo MRI scans of the abdomen.

3.2. Metrics

We adopt the metrics established in the original OpenOOD framework [72, 77], treating OOD samples as the positive class and use the area under the receiver operating characteristic curve (AUROC) as the primary metric to evaluate performance across all thresholds. It assesses the trade-off between the true positive rate (TPR) and false positive rate (FPR), measuring the model's ability to separate ID from OOD. In addition, we employ the FPR@95 metric, measuring the FPR when 95 % of the ID data is correctly classified.

We also report the area under the Precision-Recall curve (AUPR) two variants: (1) with ID samples as the positive class ($\mathrm{AUPR_{IN}}$), to measure the model's capacity to identify ID data; (2) with OOD samples as the positive class ($\mathrm{AUPR_{OUT}}$), to assess the model's ability to detect unexpected inputs. Both AUPR metrics compute precision and recall across all classification thresholds, similar to AUROC. While we present all metric results in the supplementary (Appendix B.2), we report the harmonic mean of $\mathrm{AUPR_{IN}}$ and $\mathrm{AUPR_{OUT}}$ as a single, balanced metric for our main results. To better assess model performance in imbalanced settings, we replace OpenOOD's [72, 77] accuracy metric with the F1-Score. Further details on the metric selection can be found in Appendix A.5.

4. Experiments and Results

This section begins by outlining the training pipeline for the classification models used with the three MIBs: MIDOG, PhaKIR, and OASIS-3. Subsequently, the conducted experiments are described and discussed in detail.

4.1. Classifier Training

All ID datasets are split into training, validation, and test subsets. To optimize the three MIB-specific classification models, we use the OneCycle learning rate scheduler [57] with either the SGD optimizer [63] for MIDOG and the Adam optimizer [36] for PhaKIR and OASIS-3. Final models, trained using cross-entropy loss, are selected based on the highest macro-averaged F1-score on the respective validation sets. To enhance performance and reduce training time, we initialize models with pre-trained weights without freezing any model parameters: ImageNet1k [14] for MIDOG and PhaKIR, and Kinetics400 [34] for OASIS-3. Appendix B.1 provides further details on each training setup.

4.2. Experimental Setup

We evaluate 24 post-hoc methods, initially evaluated on natural images, in the context of our MIBs, with results summarized in Tab. 1. The methods are ranked based on their average AUROC performance across all MIBs in the near-OOD setting, as this represents critical, high-risk scenarios where failure in OOD detection could lead to potential misdiagnoses or compromised patient safety.

We categorize the evaluated methods into three groups: (1) **Classification-Based Methods (blue)**: These methods primarily use information derived from the model output

Table 1. Results of all evaluated OOD detection methods, ranked by their average performance (AUROC) across all MIB groups for the **near-OOD** category. Additionally, the table includes the averaged near-OOD performance based on the harmonic mean of AUPR_{IN} and AUPR_{OUT}, as well as the averaged FPR@95 metrics. The F1-Score indicates the performance on the ID test set. Rows are color-coded to indicate the source of information used for OOD detection: blue for probabilities or logits, orange for feature space, and green for a combination of both. *: The results of the MDSEns method are potentially misleading, as discussed in the main text (Sec. 4.3).

	MIDOG @ F1-Score: 81.88			**PhaKIR** @ F1-Score: 80.08			**OASIS-3** @ F1-Score: 73.65			**Averages for near-OOD**		
	cs-ID	near-OOD	far-OOD	cs-ID	near-OOD	far-OOD	cs-ID	near-OOD	far-OOD	$\overline{\text{AUROC}}\uparrow$	$\overline{\text{AUPR}}\uparrow$	$\overline{\text{FPR@95}}\downarrow$
MDSEns* [40]	99.19	91.84	100.00	65.05	97.11	98.50	100.00	99.46	100.00	96.14	91.86	11.97
ViM [70]	59.73	62.67	84.78	72.39	81.14	55.34	98.82	98.40	100.00	80.74	70.21	48.79
Residual [70]	60.26	65.78	92.35	57.12	76.99	57.31	96.97	96.70	100.00	79.82	70.18	49.94
MDS [40]	58.60	63.21	90.91	56.04	76.48	51.47	96.31	96.15	100.00	78.61	69.72	51.29
KNN [61]	57.97	61.63	90.18	34.05	55.44	37.75	98.78	97.66	100.00	71.58	61.94	62.62
SHE [76]	57.89	61.80	91.09	36.18	50.34	47.40	91.85	90.80	99.64	67.65	58.40	67.29
RMDS [52]	49.46	52.23	60.68	38.22	67.73	35.49	58.00	71.69	99.89	63.88	48.58	79.88
Relation [35]	55.68	58.71	86.17	31.06	61.02	30.72	45.48	66.20	92.91	61.98	47.95	78.62
fDBD [44]	52.54	58.33	83.03	30.08	50.13	27.54	58.62	75.31	92.93	61.26	50.33	76.55
SCALE [71]	53.44	55.71	82.03	35.91	38.97	46.83	84.53	84.32	98.23	59.67	48.91	78.67
ReAct [60]	53.79	57.40	84.86	30.34	48.38	25.89	44.74	70.23	77.53	58.67	46.49	81.79
ASH [15]	53.93	56.02	82.70	45.99	40.17	65.08	70.71	76.52	91.48	57.57	46.13	80.71
RankFeat [58]	48.17	51.83	56.44	52.01	45.26	27.35	71.29	74.65	96.99	57.25	48.14	80.36
OpenMax [8]	49.88	52.75	65.87	31.90	66.03	33.56	43.69	52.60	70.47	57.13	40.39	90.03
ODIN [42]	56.38	61.37	85.08	34.28	41.78	71.82	53.51	59.53	58.63	54.23	44.15	86.91
GEN [46]	52.54	55.46	79.78	32.65	51.55	32.68	36.95	53.50	78.45	53.50	37.50	91.64
MSP [23]	53.37	55.90	80.91	32.04	50.16	32.51	36.95	53.50	78.45	53.19	37.29	91.71
Dropout [17]	53.25	55.76	80.87	31.96	50.10	32.56	37.10	53.22	77.13	53.03	37.17	91.88
TempScale [19]	53.57	56.05	81.38	31.73	48.73	32.50	36.95	53.50	78.45	52.76	37.05	91.65
NNGuide [49]	58.08	59.84	89.58	29.15	37.98	40.94	34.76	56.95	76.37	51.59	37.08	90.55
KLM [24]	48.34	51.54	73.53	55.15	56.34	35.87	54.51	44.49	63.98	50.79	36.57	89.40
EBO [45]	54.46	56.85	84.45	31.99	40.18	34.34	33.86	49.39	69.65	48.81	35.23	92.33
MLS [26]	54.22	56.58	82.98	32.00	40.17	34.33	33.94	49.47	69.73	48.74	35.22	92.37
DICE [59]	51.56	54.15	79.08	35.96	53.33	22.65	34.66	26.14	23.70	44.54	34.22	92.83

probabilities or logits. (2) **Feature-Based Methods (orange)**: These methods rely exclusively on information from the feature space. (3) **Hybrid Methods (green)**: These methods combine both sources of information.

For this evaluation, we extend the codebase provided by the OpenOOD benchmark [77]. To enable hyperparameter tuning, we aggregate validation splits from all near-OOD datasets within each benchmark. Fine-tuning is conducted on this combined validation set, focusing on optimizing performance in the challenging near-OOD context.

4.3. Results on all Medical Imaging Benchmarks

In this subsection we start with a description of the four top-performing methods, followed by a detailed analysis of the challenges posed by specific datasets to those methods. It also addresses the suboptimal performance of classification-based methods and compares the results with those obtained on the ImageNet1k benchmark from OpenOOD [72]. Additionally, Table 8 – Tab. 10 in the appendix provide concise summaries of all the approaches.

MDSEns and MDS The MDSEns method [40] from Lee *et al*. achieved the best overall performance across all MIBs. This method calculates the Mahalanobis distance between feature embeddings of test samples and precomputed class-conditional Gaussian distributions derived from ID training data across multiple model layers. MDSEns aggregates these distance calculations from all intermediate layers through weighted averaging, with weights optimized during an initial setup phase using logistic regression on validation data containing both ID and OOD examples.

In contrast to MDSEns, the fourth-ranked method MDS [40] computes the Mahalanobis distance using feature embeddings derived solely from the penultimate layer.

Residual and ViM Similar to MDS, the third-ranked method Residual from Wang *et al*. [70] relies entirely on features extracted from the penultimate layer. The method projects features onto a low-variance subspace defined by the N smallest eigenvalues of the empirical covariance matrix calculated from all ID training data, and uses the norm of those features as the OOD score.

ViM, an extension of the Residual method, integrates class-specific logit information to enhance OOD detection [70]. It achieves this by transforming the Residual output into a virtual logit, which is then combined with Energy-Based OOD Detection (EBO) [45]. This approach recovers information lost in the feature-to-logit mapping, consistently improving detection performance across diverse OOD scenarios with natural images.

Results on challenging datasets Figure 2 illustrates the distribution of OOD scores and corresponding AUROC values for the four top-performing methods on particularly

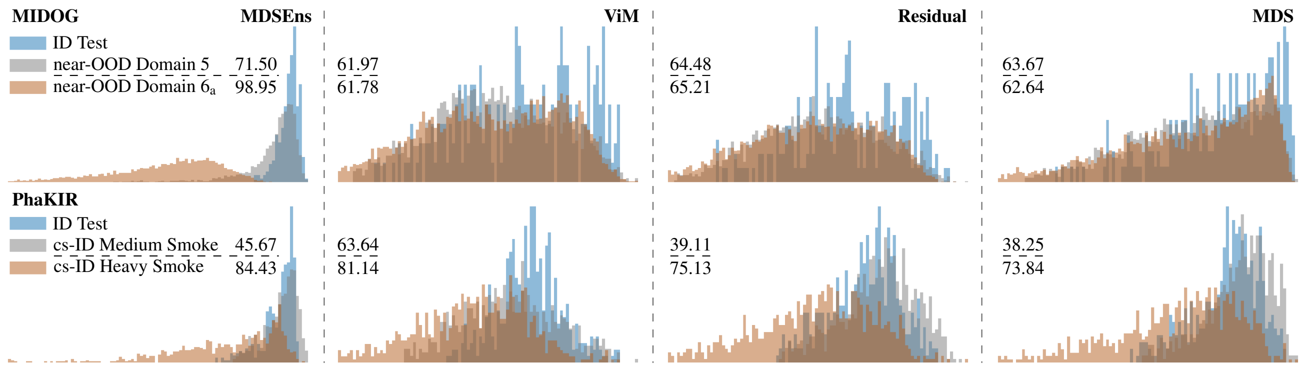

Figure 2. Distribution of OOD scores for the top four methods on challenging datasets from the MIDOG and PhaKIR benchmarks, including AUROC values for each dataset and method.

challenging datasets from MIDOG and PhaKIR. A corresponding visualization for OASIS-3 is presented in Fig. 4 in the appendix. From Tab. 1 it is evident that MDSEns delivers strong overall performance across the evaluated MIBs. However, even this top-performing method faces difficulties in reliably distinguishing between ID and OOD data under certain challenging conditions, such as given by MIDOG's domain 5 dataset and PhaKIR's Medium Smoke dataset.

This outcome aligns with the characteristics of domain 5 from MIDOG, which originates from the same scanner and institution as the ID data. Here, the primary shift is semantic, involving a different cell type from the same species. In contrast, domain 6_a exhibits both a more pronounced semantic shift – a different cell type from another species – and additional covariate shifts introduced by variations in scanner type and institution. While MDSEns achieves near-perfect separation in domain 6_a, where both semantic and covariate shifts are present, it degrades significantly in domain 5, where only a semantic shift is involved.

This finding indicates that MDSEns' strong performance is heavily influenced by its ability to detect covariate shifts, which are often more prominent in the datasets used. MDSEns uniquely uses information from early intermediate layers of the network, where covariate shifts are more likely to manifest as changes in basic features like edges, shapes, and colors [37, 75]. This observation aligns with prior findings [68], demonstrating that simple approaches based on input intensity distributions can yield results comparable to methods that rely on deeper network layers or final outputs.

For the PhaKIR Medium Smoke dataset, the overlap between OOD and ID scores can be attributed to the subtle appearance of smoke in the images. Often, the smoke is limited to regions that do not contain instruments. Given that the model's primary task is instrument classification, it likely prioritizes regions containing instruments over background areas. As a result, features associated with smoke, particularly in non-instrument regions, are minimally encoded. A representative example of this behavior is shown in Fig. 5 in the appendix. If the downstream task were instead organ classification, it is likely that smoke would have a stronger impact on the encoded features, as it could obscure or alter key characteristics of the organs themselves.

Comparing the performance of ViM and Residual [70] on PhaKIR's cs-ID Medium Smoke dataset reveals a significant performance decline for Residual (Fig. 2). ViM differs from Residual primarily in its computation of a virtual-logit and its integration with the EBO method. Given EBO's subpar performance on this dataset (23.74, Tab. 14 located in the appendix), ViM's improvement is likely attributed to the virtual-logit calculation, which seeks to recover lost information during the feature-to-logit mapping.

Poor performance of classification-based methods An evaluation of the average AUROC for classification-based approaches across the three MIBs reveals a significant lack of discriminative power compared to feature-based methods (Tab. 7 in the appendix). These approaches inherently rely on the assumption that ID data should yield higher confidence scores than OOD data. However, this assumption is often violated in practice, as demonstrated by our analysis.

For example, in the PhaKIR benchmark, the mean softmax probability for ID predictions of the "Grasper" class is 79.07 %, whereas for OOD predictions from the EndoSeg18 dataset, the mean probability on all Grasper predictions increases to 94.04 %. This paradoxical result indicates a critical flaw: the classifier overconfidently assigns the Grasper label to OOD data. Analyzing the feature space reveals that OOD data from EndoSeg18 clusters predominantly near the Grasper class (Fig. 6 in the appendix), suggesting a bias in the classifier toward assigning this label to instruments that do not clearly belong to any other class. Together, these factors explain the low average AUROC of 33.71 for classification-based methods on the EndoSeg18 dataset.

Similar patterns of poor discrimination are observed

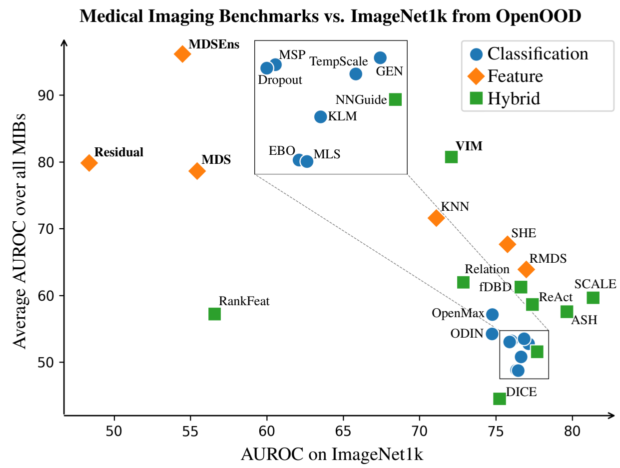

Figure 3. Average ranking of each method based on AUROC across the MIBs (y-axis) and on the ImageNet1k benchmark (x-axis).

across other datasets. In the Kvasir-SEG dataset, for example, the classifier assigned higher scores to predictions for the ”No-Instrument” class than it did for ID data. Similarly, in the OASIS-3 benchmark, outputs for the CN class are nearly identical between the ID and the ATLAS dataset, compromising the ability of classification-based methods to distinguish between ID and OOD samples in each case.

Comparison with OpenOOD’s ImageNet1k benchmark Figure 3 compares the average AUROC across the MIBs for all evaluated methods with their performance on the ImageNet1k benchmark from OpenOOD [72]. The results highlight that no single method fully addresses the challenge of OOD detection across both natural and medical imaging scenarios. The closest contender of a comprehensive approach is ViM, which combines information from both feature space and the classification layers. ViM achieves an average AUROC of 80.74 on the MIBs and 72.08 on the ImageNet1k benchmark, demonstrating its adaptability to different domains. However, the performance gap between these benchmarks underscores the complexity and specificity of OOD detection in medical imaging, where domain characteristics significantly differ from those of natural images. Notably, methods that rely primarily on feature space information tend to perform better on medical images, while methods leveraging probabilities or logits show superior results on natural images. This discrepancy may stem from the lower variance typically observed in medical images compared to natural images. For instance, the average standard deviation in pixel intensity is 0.148/0.149 for the MIDOG/PhaKIR ID datasets, respectively, compared to 0.226 for the ImageNet1k dataset. The lower variance in medical images, along with the typically small number of target classes in medical datasets, may result in more cohesive and less fragmented feature spaces, thereby enhancing the effectiveness of feature-based methods for detecting OOD samples, presenting a starting point for future research. On the other hand, natural images, with their higher variability and richer textures, likely benefit from methods that exploit class probabilities and logits. Detailed results on ImageNet1k can be found in Appendix B.2.

5. Conclusion

In this work, we introduced the Open Medical Imaging Benchmarks for Out-Of-Distribution Detection (OpenMIBOOD), representing a significant step forward in the evaluation and development of OOD detection methods for medical imaging. OpenMIBOOD comprises a comprehensive set of three benchmarks consisting of 14 datasets, categorized into in-distribution, covariate-shifted in-distribution, near-OOD, and far-OOD. Across these benchmarks, we evaluated 24 post-hoc OOD detection methods and analyzed their performance and limitations in the medical field.

Our results reveal that state-of-the-art OOD detection methods that rely on feature space information consistently outperform methods that depend solely on probabilities and logits when applied to medical datasets. However, our findings also highlight a critical challenge: methods originally designed and optimized for natural images often fail to generalize to medical data. This underscores the need for tailored solutions that address the unique characteristics of medical imaging, such as lower variance, domain-specific semantic shifts, and imbalanced datasets. The precise reasons behind these discrepancies remain an area for further research. One notable limitation of our benchmark is its focus on classification tasks, which limits the applicability of our findings to such problems. Future work could extend the evaluation framework to seamlessly include segmentation tasks, a critical component of medical imaging, by adapting the included OOD detection methods to operate in segmentation settings. While we focus on evaluating OOD detection methods using deep Neural Networks, we recognize the potential of other approaches. For example, analyzing and evaluating input data distributions and directly comparing them to training set distributions could complement Neural Network-based OOD detection.

As Artificial Intelligence continues to advance in the medical field, the insights and benchmarks provided by OpenMIBOOD will play a vital role in driving innovation. By exposing the strength and weaknesses of existing methods and setting a high standard for future development, OpenMIBOOD contributes to the creation of trustworthy AI systems. These systems will be essential for safely and effectively addressing the challenges of clinical application, ensuring that AI technologies enhance patient care while maintaining high standards of reliability and safety.

References

[1] Max Allan, Satoshi Kondo, Sebastian Bodenstedt, Stefan Leger, Rahim Kadkhodamohammadi, Imanol Luengo, Felix Fuentes, Evangello Flouty, Ahmed Mohammed, Marius Pedersen, et al. 2018 Robotic Scene Segmentation Challenge. *ArXiv preprint*, abs/2001.11190, 2020. 2, 4

[2] João Gustavo Atkinson Amorim, Luiz Antonio Buschetto Macarini, André Victória Matias, Allan Cerentini, Fabiana Botelho De Miranda Onofre, Alexandre Sherlley Casimiro Onofre, and Aldo Von Wangenheim. A novel approach on segmentation of agnor-stained cytology images using deep learning. In *Proceedings of the 2020 IEEE 33rd International Symposium on Computer-Based Medical Systems (CBMS)*, pages 552–557. IEEE, 2020. 2, 4, 1

[3] Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A Landman, Geert Litjens, Bjoern Menze, Olaf Ronneberger, Ronald M Summers, et al. The medical segmentation decathlon. *Nature communications*, 13(1):4128, 2022. 5, 3

[4] João Gustavo Atkinson Amorim, André Matias, Tainee Bottamedi, Vinícius Sanches, Ane Francyne Costa, Fabiana Onofre, Alexandre Onofre, and Aldo Wangenheim. CCAgT: Images of Cervical Cells with AgNOR Stain Technique, 2022. Accessed: 2024-10-13, Publisher: Mendeley Data. 2, 4, 1

[5] Marc Aubreville, Frauke Wilm, Nikolas Stathonikos, Katharina Breininger, Taryn A Donovan, Samir Jabari, Mitko Veta, Jonathan Ganz, Jonas Ammeling, Paul J van Diest, et al. A comprehensive multi-domain dataset for mitotic figure detection. *Scientific data*, 10(1):484, 2023. 2, 3, 1

[6] Ujjwal Baid, Satyam Ghodasara, Suyash Mohan, Michel Bilello, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C Kitamura, Sarthak Pati, et al. The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification. *ArXiv preprint*, abs/2107.02314, 2021. 5, 3

[7] Spyridon Bakas, Hamed Akbari, Aristeidis Sotiras, Michel Bilello, Martin Rozycki, Justin S Kirby, John B Freymann, Keyvan Farahani, and Christos Davatzikos. Advancing the Cancer Genome Atlas Glioma MRI Collections with Expert Segmentation Labels and Radiomic Features. *Scientific data*, 4(1):1–13, 2017. 5, 3

[8] Abhijit Bendale and Terrance E. Boult. Towards Open Set Deep Networks. In *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016, Las Vegas, NV, USA, June 27-30, 2016*, pages 1563–1572. IEEE Computer Society, 2016. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[9] Julian Bitterwolf, Maximilian Müller, and Matthias Hein. In or Out? Fixing ImageNet Out-of-Distribution Detection Evaluation. In *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, pages 2471–2506. PMLR, 2023. 16, 17

[10] Sebastian Bodenstedt, Max Allan, Anthony Agustinos, Xiaofei Du, Luis Garcia-Peraza-Herrera, Hannes Kenngott, Thomas Kurmann, Beat Müller-Stich, Sebastien Ourselin, Daniil Pakhomov, et al. Comparative Evaluation of Instrument Segmentation and Tracking Methods in Minimally Invasive Surgery. *ArXiv preprint*, abs/1805.02475, 2018. 2, 4

[11] Daniel Bogdoll, Maximilian Nitsche, and J. Marius Zöllner. Anomaly Detection in Autonomous Driving: A Survey. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPR Workshops 2022, New Orleans, LA, USA, June 19-20, 2022*, pages 4487–4498. IEEE, 2022. 1

[12] Tianshi Cao, Chin-Wei Huang, David Yu-Tung Hui, and Joseph Paul Cohen. A benchmark of medical out of distribution detection. *ArXiv preprint*, abs/2007.04250, 2020. 2, 3

[13] Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing Textures in the Wild. In *2014 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2014, Columbus, OH, USA, June 23-28, 2014*, pages 3606–3613. IEEE Computer Society, 2014. 16, 17

[14] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Fei-Fei Li. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2009), 20-25 June 2009, Miami, Florida, USA*, pages 248–255. IEEE Computer Society, 2009. 5

[15] Andrija Djurisic, Nebojsa Bozanic, Arjun Ashok, and Rosanne Liu. Extremely Simple Activation Shaping for Out-of-Distribution Detection. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[16] Zhen Fang, Yixuan Li, Jie Lu, Jiahua Dong, Bo Han, and Feng Liu. Is Out-of-Distribution Detection Learnable? In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, 2022. 1

[17] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. In *Proceedings of the 33nd International Conference on Machine Learning, ICML 2016, New York City, NY, USA, June 19-24, 2016*, pages 1050–1059. JMLR.org, 2016. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[18] Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and Harnessing Adversarial Examples. In *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015. 1

[19] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, pages 1321–1330. PMLR, 2017. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[20] Hassan Al Hajj, Mathieu Lamard, Pierre-Henri Conze, Soumali Roychowdhury, Xiaowei Hu, Gabija Marsalkaite, Odysseas Zisimopoulos, Muneer Ahmad Dedmari, Fenqiang Zhao, Jonas Prellberg, Manish Sahu, Adrian Galdran, Teresa

Araújo, Duc My Vo, Chandan Panda, Navdeep Dahiya, Satoshi Kondo, Zhengbing Bian, and Gwenolé Quellec. CATARACTS: challenge on automatic tool annotation for cataract surgery. *Medical Image Anal.*, 52:24–41, 2019. 2, 4

[21] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016, Las Vegas, NV, USA, June 27-30, 2016*, pages 770–778. IEEE Computer Society, 2016. 5

[22] Dan Hendrycks and Thomas G. Dietterich. Benchmarking Neural Network Robustness to Common Corruptions and Perturbations. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019. 16, 17

[23] Dan Hendrycks and Kevin Gimpel. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. In *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*. OpenReview.net, 2017. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[24] Dan Hendrycks, Steven Basart, Mantas Mazeika, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. A Benchmark for Anomaly Segmentation. *ArXiv preprint*, abs/1911.11132, 2019. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[25] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, Dawn Song, Jacob Steinhardt, and Justin Gilmer. The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization. In *2021 IEEE/CVF International Conference on Computer Vision, ICCV 2021, Montreal, QC, Canada, October 10-17, 2021*, pages 8320–8329. IEEE, 2021. 16, 17

[26] Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joseph Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling Out-of-Distribution Detection for Real-World Settings. In *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, pages 8759–8773. PMLR, 2022. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[27] Zesheng Hong, Yubiao Yue, Yubin Chen, Huanjie Lin, Yuanmei Luo, Mini Han Wang, Weidong Wang, Jialong Xu, Xiaoqi Yang, Zhenzhang Li, et al. Out-of-distribution Detection in Medical Image Analysis: A survey. *ArXiv preprint*, abs/2404.18279, 2024. 2, 3

[28] Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alexander Shepard, Hartwig Adam, Pietro Perona, and Serge J. Belongie. The INaturalist Species Classification and Detection Dataset. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*, pages 8769–8778. IEEE Computer Society, 2018. 16, 17

[29] Fabian Isensee, Marianne Schell, Irada Pflueger, Gianluca Brugnara, David Bonekamp, Ulf Neuberger, Antje Wick, Heinz-Peter Schlemmer, Sabine Heiland, Wolfgang Wick, et al. Automated brain extraction of multisequence MRI using artificial neural networks. *Human brain mapping*, 40 (17):4952–4964, 2019. 5, 3

[30] ISO. *ISO/IEC TS5723:2022: Trustworthiness — Vocabulary*. International Organization for Standardization, Geneva, Switzerland, 2022. 1

[31] Debesh Jha, Pia H Smedsrud, Michael A Riegler, Pål Halvorsen, Thomas De Lange, Dag Johansen, and Håvard D Johansen. Kvasir-seg: A segmented polyp dataset. In *MultiMedia modeling: 26th international conference, MMM 2020, Daejeon, South Korea, January 5–8, 2020, proceedings, part II 26*, pages 451–462. Springer, 2020. 2, 4

[32] Ali Emre Kavur, M. Alper Selver, Oğuz Dicle, Mustafa Barış, and N. Sinem Gezer. CHAOS - Combined (CT-MR) Healthy Abdominal Organ Segmentation Challenge Data, 2019. 2, 5, 3

[33] A. Emre Kavur, Naciye Sinem Gezer, Mustafa Baris, Sinem Aslan, Pierre-Henri Conze, Vladimir Groza, Duc Duy Pham, Soumick Chatterjee, Philipp Ernst, Savas Özkan, Bora Baydar, Dmitry A. Lachinov, Shuo Han, Josef Pauli, Fabian Isensee, Matthias Perkonigg, Rachana Sathish, Ronnie Rajan, Debdoot Sheet, Gurbandurdy Dovletov, Oliver Speck, Andreas Nürnberger, Klaus H. Maier-Hein, Gözde B. Akar, Gözde B. Ünal, Oguz Dicle, and M. Alper Selver. CHAOS Challenge - combined (CT-MR) healthy abdominal organ segmentation. *Medical Image Anal.*, 69:101950, 2021. 5, 3

[34] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. The kinetics human action video dataset. *ArXiv preprint*, abs/1705.06950, 2017. 5

[35] Jang-Hyun Kim, Sangdoo Yun, and Hyun Oh Song. Neural Relation Graph: A Unified Framework for Identifying Label Noise and Outlier Data. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[36] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015. 5

[37] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. Imagenet Classification with Deep Convolutional Neural Networks. In *Advances in Neural Information Processing Systems 25: 26th Annual Conference on Neural Information Processing Systems 2012. Proceedings of a meeting held December 3-6, 2012, Lake Tahoe, Nevada, United States*, pages 1106–1114, 2012. 7

[38] Pamela J LaMontagne, Tammie LS Benzinger, John C Morris, Sarah Keefe, Russ Hornbeck, Chengjie Xiong, Elizabeth Grant, Jason Hassenstab, Krista Moulder, Andrei G Vlassenko, et al. Oasis-3: longitudinal neuroimaging, clinical, and cognitive dataset for normal aging and Alzheimer disease. *medrxiv*, pages 2019–12, 2019. 2, 4, 3

[39] Yann LeCun, Sumit Chopra, Raia Hadsell, M Ranzato, Fujie Huang, et al. A Tutorial on Energy-Based Learning. *Predicting structured data*, 1(0), 2006. 8

[40] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A Simple Unified Framework for Detecting Out-of-

Distribution Samples and Adversarial Attacks. In *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, pages 7167–7177, 2018. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[41] Robert M Levy, Catherine M Mills, Jonathan P Posin, Sheila G Moore, Mark L Rosenblum, and Dale E Bredesen. The efficacy and clinical impact of brain imaging in neurologically symptomatic AIDS patients: a prospective CT/MRI study. *JAIDS Journal of Acquired Immune Deficiency Syndromes*, 3(5):461–471, 1990. 5

[42] Shiyu Liang, Yixuan Li, and R. Srikant. Enhancing The Reliability of Out-of-distribution Image Detection in Neural Networks. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[43] Sook-Lei Liew, Bethany P Lo, Miranda R Donnelly, Artemis Zavaliangos-Petropulu, Jessica N Jeong, Giuseppe Barisano, Alexandre Hutton, Julia P Simon, Julia M Juliano, Anisha Suri, et al. A large, curated, open-source stroke neuroimaging dataset to improve lesion segmentation algorithms. *Scientific data*, 9(1):320, 2022. 2, 5, 3

[44] Litian Liu and Yao Qin. Fast Decision Boundary based Out-of-Distribution Detector. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[45] Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based Out-of-distribution Detection. In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[46] Xixi Liu, Yaroslava Lochman, and Christopher Zach. GEN: Pushing the Limits of Softmax-Based Out-of-Distribution Detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, pages 23946–23955. IEEE, 2023. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[47] Bjoern H. Menze, András Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin S. Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, Levente Lanczi, Elizabeth R. Gerstner, Marc-André Weber, Tal Arbel, Brian B. Avants, Nicholas Ayache, Patricia Buendia, D. Louis Collins, Nicolas Cordier, Jason J. Corso, Antonio Criminisi, Tilak Das, Herve Delingette, Çagatay Demiralp, Christopher R. Durst, Michel Dojat, Senan Doyle, Joana Festa, Florence Forbes, Ezequiel Geremia, Ben Glocker, Polina Golland, Xiaotao Guo, Andac Hamamci, Khan M. Iftekharuddin, Raj Jena, Nigel M. John, Ender Konukoglu, Danial Lashkari, José Antonio Mariz, Raphael Meier, Sérgio Pereira, Doina Precup, Stephen J. Price, Tammy Riklin Raviv, Syed M. S. Reza, Michael T. Ryan, Duygu Sarikaya, Lawrence H. Schwartz, Hoo-Chang Shin, Jamie Shotton, Carlos A. Silva, Nuno J. Sousa, Nagesh K. Subbanna, Gábor Székely, Thomas J. Taylor, Owen M. Thomas, Nicholas J. Tustison, Gözde B. Ünal, Flor Vasseur, Max Wintermark, Dong Hye Ye, Liang Zhao, Binsheng Zhao, Darko Zikic, Marcel Prastawa, Mauricio Reyes, and Koen Van Leemput. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE Trans. Medical Imaging*, 34(10):1993–2024, 2015. 2, 5, 3

[48] Samuel G. Müller and Frank Hutter. Trivialaugment: Tuning-free Yet State-of-the-Art Data Augmentation. In *2021 IEEE/CVF International Conference on Computer Vision, ICCV 2021, Montreal, QC, Canada, October 10-17, 2021*, pages 754–762. IEEE, 2021. 5

[49] Jaewoo Park, Yoon Gyo Jung, and Andrew Beng Jin Teoh. Nearest Neighbor Guidance for Out-of-Distribution Detection. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, pages 1686–1695. IEEE, 2023. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[50] Konstantin Pogorelov, Kristin Ranheim Randel, Carsten Griwodz, Sigrun Losada Eskeland, Thomas de Lange, Dag Johansen, Concetto Spampinato, Duc-Tien Dang-Nguyen, Mathias Lux, Peter Thelin Schmidt, et al. Kvasir: A multi-class image dataset for computer aided gastrointestinal disease detection. In *Proceedings of the 8th ACM on Multimedia Systems Conference*, pages 164–169, 2017. 2

[51] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet Classifiers Generalize to ImageNet? In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, pages 5389–5400. PMLR, 2019. 16, 17

[52] Jie Ren, Stanislav Fort, Jeremiah Liu, Abhijit Guha Roy, Shreyas Padhy, and Balaji Lakshminarayanan. A Simple Fix to Mahalanobis Distance for Improving Near-OOD Detection. *ArXiv preprint*, abs/2106.09022, 2021. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[53] Tobias Rueckert. PhaKIR-Challenge at MICCAI 2024. https://phakir.re-mic.de/, 2024. Accessed: 2024-10-13. 2, 4, 1

[54] Tobias Rueckert, Maximilian Rieder, David Rauber, Michel Xiao, Eg Humolli, Hubertus Feussner, Dirk Wilhelm, and Christoph Palm. Augmenting instrument segmentation in video sequences of minimally invasive surgery by synthetic smoky frames. In *International Journal of Computer Assisted Radiology and Surgery*, pages S54 – S56, 2023. 4, 1

[55] Amartya Ranjan Saikia, Kangkana Bora, Lipi B Mahanta, and Anup Kumar Das. Comparative assessment of CNN architectures for classification of breast FNAC images. *Tissue and Cell*, 57:8–14, 2019. 2, 4, 1

[56] Amber L Simpson, Michela Antonelli, Spyridon Bakas, Michel Bilello, Keyvan Farahani, Bram Van Ginneken, Annette Kopp-Schneider, Bennett A Landman, Geert Litjens, Bjoern Menze, et al. A large annotated medical image dataset for the development and evaluation of segmentation algorithms. *ArXiv preprint*, abs/1902.09063, 2019. 5

[57] Leslie N Smith and Nicholay Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial intelligence and machine learning*

for multi-domain operations applications, pages 369–386. SPIE, 2019. 5

[58] Yue Song, Nicu Sebe, and Wei Wang. RankFeat: Rank-1 Feature Removal for Out-of-distribution Detection. In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, 2022. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[59] Yiyou Sun and Yixuan Li. DICE: Leveraging Sparsification for Out-of-Distribution Detection. In *Computer Vision - ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23-27, 2022, Proceedings, Part XXIV*, pages 691–708. Springer, 2022. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[60] Yiyou Sun, Chuan Guo, and Yixuan Li. ReAct: Out-of-distribution Detection With Rectified Activations. In *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 144–157, 2021. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[61] Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-Distribution Detection with Deep Nearest Neighbors. In *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, pages 20827–20840. PMLR, 2022. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[62] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic Attribution for Deep Networks. In *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, pages 3319–3328. PMLR, 2017. 6

[63] Ilya Sutskever, James Martens, George E. Dahl, and Geoffrey E. Hinton. On the importance of initialization and momentum in deep learning. In *Proceedings of the 30th International Conference on Machine Learning, ICML 2013, Atlanta, GA, USA, 16-21 June 2013*, pages 1139–1147. JMLR.org, 2013. 5

[64] Catalina Tobon-Gomez, Arjan J. Geers, Jochen Peters, Jürgen Weese, Karen Pinto, Rashed Karim, Mohammed Ammar, Abdelaziz Daoudi, Ján Margeta, Zulma L. Sandoval, Birgit Stender, Yefeng Zheng, Maria A. Zuluaga, Julián Betancur, Nicholas Ayache, Mohammed Amine Chikh, Jean-Louis Dillenseger, B. Michael Kelm, Saïd Mahmoudi, Sébastien Ourselin, Alexander Schlaefer, Tobias Schaeffter, Reza Razavi, and Kawal S. Rhode. Benchmark for Algorithms Segmenting the Left Atrium From 3D CT and MRI Datasets. *IEEE Trans. Medical Imaging*, 34(7):1460–1473, 2015. 2, 5, 3

[65] Du Tran, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri. A Closer Look at Spatiotemporal Convolutions for Action Recognition. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*, pages 6450–6459. IEEE Computer Society, 2018. 5

[66] Andru Putra Twinanda, Sherif Shehata, Didier Mutter, Jacques Marescaux, Michel de Mathelin, and Nicolas Padoy. Endonet: A Deep Architecture for Recognition Tasks on Laparoscopic Videos. *IEEE Trans. Medical Imaging*, 36(1): 86–97, 2017. 2, 4

[67] Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of machine learning research*, 9 (11), 2008. 6

[68] Anton Vasiliuk, Daria Frolova, Mikhail Belyaev, and Boris Shirokikh. Limitations of Out-of-Distribution Detection in 3D Medical Image Segmentation. *Journal of Imaging*, 9(9): 191, 2023. 2, 3, 7

[69] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-Set Recognition: A Good Closed-Set Classifier is All You Need. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022. 16, 17

[70] Haoqi Wang, Zhizhong Li, Litong Feng, and Wayne Zhang. ViM: Out-Of-Distribution with Virtual-logit Matching. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pages 4911–4920. IEEE, 2022. 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17

[71] Kai Xu, Rongyu Chen, Gianni Franchi, and Angela Yao. Scaling for Training Time and Post-hoc Out-of-distribution Detection Enhancement. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. 6, 9, 10, 11, 12, 13, 14, 15, 16, 17

[72] Jingkang Yang, Pengyun Wang, Dejian Zou, Zitang Zhou, Kunyuan Ding, Wenxuan Peng, Haoqi Wang, Guangyao Chen, Bo Li, Yiyou Sun, Xuefeng Du, Kaiyang Zhou, Wayne Zhang, Dan Hendrycks, Yixuan Li, and Ziwei Liu. OpenOOD: Benchmarking Generalized Out-of-Distribution Detection. In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, 2022. 1, 2, 3, 5, 6, 8, 7

[73] Jingkang Yang, Kaiyang Zhou, and Ziwei Liu. Full-Spectrum Out-of-Distribution Detection. *International Journal of Computer Vision*, 131(10):2607–2622, 2023. 1, 6

[74] Jingkang Yang, Kaiyang Zhou, Yixuan Li, and Ziwei Liu. Generalized Out-of-Distribution Detection: A Survey. *International Journal of Computer Vision*, pages 1–28, 2024. 1, 3

[75] Matthew D. Zeiler and Rob Fergus. Visualizing and Understanding Convolutional Networks. In *Computer Vision - ECCV 2014 - 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part I*, pages 818–833. Springer, 2014. 7

[76] Jinsong Zhang, Qiang Fu, Xu Chen, Lun Du, Zelin Li, Gang Wang, Xiaoguang Liu, Shi Han, and Dongmei Zhang. Out-of-Distribution Detection based on In-Distribution Data Patterns Memorization with Modern Hopfield Energy. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. 6, 8, 10, 11, 12, 13, 14, 15, 16, 17

[77] Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yueqian Lin, Haoran Zhang, Yiyou Sun, Xuefeng Du, Kaiyang Zhou, Wayne Zhang, Yixuan Li, Ziwei Liu, Yiran Chen, and Hai Li. OpenOOD v1.5: Enhanced Benchmark

for Out-of-Distribution Detection. In *NeurIPS 2023 Workshop on Distribution Shifts: New Frontiers with Foundation Models*, 2024. 1, 2, 3, 5, 6, 7, 9

[78] Wen-zhen Zhu, Jian-pin Qi, Chuan-jia Zhan, Hong-ge Shu, Lin Zhang, Cheng-yuan Wang, Li-ming Xia, Jun-wu Hu, and Ding-yi Feng. Magnetic resonance susceptibility weighted imaging in detecting intracranial calcification and hemorrhage. *Chinese medical journal*, 121(20):2020–2025, 2008. 5

[79] David Zimmerer, Peter M. Full, Fabian Isensee, Paul Jäger, Tim Adler, Jens Petersen, Gregor Köhler, Tobias Roß, Annika Reinke, Antanas Kascenas, Bjørn Sand Jensen, Alison Q. O'Neil, Jeremy Tan, Benjamin Hou, James Batten, Huaqi Qiu, Bernhard Kainz, Nina Shvetsova, Irina Fedulova, Dmitry V. Dylov, Baolun Yu, Jianyang Zhai, Jingtao Hu, Runxuan Si, Sihang Zhou, Siqi Wang, Xinyang Li, Xuerun Chen, Yang Zhao, Sergio Naval Marimont, Giacomo Tarroni, Victor Saase, Lena Maier-Hein, and Klaus H. Maier-Hein. MOOD 2020: A Public Benchmark for Out-of-Distribution Detection and Localization on Medical Images. *IEEE Trans. Medical Imaging*, 41(10):2728–2738, 2022. 2